

S4D-TAM: Semantic 4D Token Attention Map for Autonomous Navigation of Unmanned Aerial Vehicles in GNSS-Degraded Environments

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Abstract

We propose S4D-TAM (Semantic 4D Token Attention Map), a novel navigation system architecture for unmanned aerial vehicles (UAVs) operating in GNSS-degraded environments. Unlike classical methods that repeatedly analyze raw image streams, the proposed system transforms sensor data (RGB, thermal, IMU, GNSS) into a persistent map of semantic 4D tokens. Each token represents a spatial fragment or object instance and contains its geometry, semantic embedding, dynamics, observation history, and calibrated uncertainty. A hierarchical spatiotemporal transformer updates the world model, aligns it with reference maps, and forecasts occupancy in partially observable regions. The predictive map is then used by a planner that accounts for risk, energy cost, and information value of future observations. The proposed representation aims to reduce image processing redundancy, increase robustness to GNSS degradation, and enable decision-making based on persistent spatial memory rather than solely on the current sensor field of view. The article presents the system concept, literature review, mathematical description, an executable auditable reference implementation, and a two-level experimental methodology separating external system comparison from internal mechanism ablation. Confirmatory experimental results are not yet reported.

Keywords: unmanned aerial vehicles, semantic SLAM, 4D tokens, spatiotemporal transformer, autonomous navigation, GNSS degradation, world model, occupancy forecasting

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1. Introduction

Autonomous navigation of unmanned aerial vehicles (UAVs) in environments with limited GNSS signal availability is one of the central challenges of modern mobile robotics. Satellite signal degradation can occur in dense urban environments, under heavy forest canopy, in tunnels and canyons, and under deliberate jamming or spoofing conditions [19], [20], [47], [48]. In such scenarios, a UAV must rely on alternative information sources: RGB and thermal cameras, inertial measurement units (IMU), altimeters, LiDARs, and reference maps [26], [53], [54], [57].

Contemporary autonomous navigation systems perform pose estimation, map construction, semantic segmentation, object recognition, occupancy forecasting, and trajectory planning as separate, loosely coupled modules [9], [31]. This approach leads to computational redundancy - the same images are repeatedly analyzed by different modules, and semantic and geometric information is not efficiently shared between perception and planning stages [12], [40].

In this work, we propose S4D-TAM (Semantic 4D Token Attention Map), a novel navigation system architecture in which the fundamental environment representation is a persistent, semantic, continuously updated map of 4D tokens derived from the SLAM point cloud. RGB and thermal camera data are used primarily to update the geometry and semantic meaning of points and objects stored in the map. Planning, risk assessment, and decision-making modules do not repeatedly analyze the same images - they operate on a compressed spatial representation containing geometry, semantics, dynamics, observation history, and uncertainty [1].

The key scientific novelty of the proposed system is the introduction of a persistent memory map composed of semantic 4D tokens, which serves as a unified representation for localization, environment understanding, future scene state prediction, risk estimation, and UAV trajectory planning. The system remembers not only where an object was located, but also when and how many times it was observed, from which sensors the information originated, how reliable its position and recognition are, whether the object is static, dynamic, or periodically displaced,

what spatial relations hold between it and other terrain elements, what invisible areas may be hidden behind it, and what significance the object has for flight safety and planning [1].

The article combines a conceptual system architecture with an executable research reference and a reproducible benchmark framework. The current repository implements token state and lifecycle management, proposal and association, multimodal encoder interfaces, attention-related processing, uncertainty calibration utilities, reference-map and topological support, causal occupancy and motion forecasting, deterministic risk-aware planning, telemetry, and the integrated S4DTAMReference execution path. The final learned hierarchical model, full public-dataset reproduction, and confirmatory results remain under development. Section 7 therefore specifies a prospective two-level validation protocol rather than reporting final performance results.

2. Literature Review

2.1. Semantic SLAM for UAVs

Semantic SLAM for aerial platforms must be lightweight, robust to GPS signal loss, and capable of producing useful semantic maps for planning and multi-robot coordination. Liu et al. proposed SlideSLAM - a sparse, lightweight, decentralized metric-semantic SLAM for multi-robot navigation that uses an object-based representation and enables efficient map merging in heterogeneous aerial-ground teams [2]. Tao et al. integrated active exploration with metric-semantic SLAM for resource-constrained aerial robots, explicitly balancing exploration and localization uncertainty (including semantic loop closures), which informs S4D-TAM strategies for uncertainty-aware mapping in GNSS-denied environments [3]. Liu et al. presented a real-time semantic SLAM that detects and models tree trunks and planes from LiDAR to constrain poses and enable drift compensation during long-range autonomous flights under forest canopy, directly relevant to drift mitigation and semantic constraints in S4D-TAM trajectory planning [4].

Tian et al. proposed SAME - a ground-air collaboration system that fuses semantic Octree units from a UAV and a UGV into a coherent global map supporting collaborative exploration, offering a semantic octree architecture and fusion approach useful in S4D-TAM for cross-platform mapping and multi-perspective scene reconstruction [6]. Xiang and Su developed a LiDAR-based semantic mapping pipeline that creates hierarchical semantic maps together with occupancy grids for navigation and task adaptation, illustrating practical semantic mapping designs and hierarchical semantics that S4D-TAM can exploit for task-oriented scene understanding [7].

Li et al. introduced Hi-SLAM - a hierarchical categorical representation within 3D Gaussian Splatting for compact encoding of multiple semantic classes and improved semantic mapping and tracking, directly applicable to S4D-TAM requirements for scalable semantic scene representations [5], [46]. More recent semantic SLAM works employing 3D Gaussian Splatting include SGS-SLAM [27], GS4 [28], SemGauss-SLAM [29], SAP-SLAM [30], DynaGS-SLAM [32], GSFF-SLAM [33], OVO-SLAM [34], MSGS-SLAM [35], NIS-SLAM [36], and NEDS-SLAM [45], demonstrating growing interest in real-time dense semantic representations for SLAM.

2.2. Point Cloud Transformers

Point cloud transformer architectures and related models provide expressive spatiotemporal encoding of 3D geometry and dynamics, useful for the point-cloud-based perception and temporal prediction modules of S4D-TAM. PCPNet introduces a transformer network for future LiDAR point cloud prediction with a semantic auxiliary training objective for

semantically consistent forecasts, directly applicable to S4D-TAM for point-level prediction and semantic alignment [16]. Silva et al. proposed S2TPVFormer, which extends TPV embeddings with hybrid cross-view temporal attention for temporally consistent semantic 3D occupancy, offering a spatiotemporal transformer design that S4D-TAM can adopt for temporally consistent semantic occupancy mapping [12].

Chen et al. presented AdaOcc, which employs adaptive forward-view transformations and flow modeling (with strong Swin camera backbone models) to improve voxel occupancy and flow prediction, providing practical camera-to-3D transformer strategies applicable in vision-oriented S4D-TAM modules [15]. Work on point cloud forecasting as a proxy for 4D occupancy prediction redefines the problem toward sensor-agnostic spatiotemporal (4D) occupancy forecasting, directing S4D-TAM toward world-oriented 4D occupancy targets for robust temporal modeling [8].

Further relevant works include Point-BERT [43], which introduces masked point modeling for pre-training point cloud transformers; SAT3D [42] for semantic segmentation of 3D scenes using slot attention; and YOGO [37] and Efficient Point Transformer [38], [39], demonstrating efficient point cloud processing with token representation and relation inference modules relevant to S4D-TAM token compression.

2.3. 4D Occupancy Forecasting

Accurate 4D occupancy forecasting provides the spatiotemporal scene model required by S4D-TAM for risk assessment and trajectory planning. Yu et al. proposed Drive-OccWorld, which adapts a vision-centric 4D world model with memory normalization and action-conditioned decoding for occupancy and flow prediction, and integrates forecasting with end-to-end planning, providing an approach that S4D-TAM can leverage to couple occupancy prediction with trajectory selection [8]. Zheng et al. presented OccWorld - a spatiotemporal generative transformer architecture for 4D occupancy prediction, demonstrating effectiveness on driving datasets and offering S4D-TAM a transformer backbone for long-horizon volumetric prediction [3].

Ma et al. provided Cam4DOcc - a standardized camera-only benchmark and four baselines for current and future occupancy estimation, providing S4D-TAM with evaluation protocols and camera-only baseline designs for 4D occupancy forecasting [4]. Guo et al. proposed FSF-Net, which uses coarse BEV scene flow plus vector-quantized latent features and U-Net fusion to improve occupancy prediction accuracy, showing a practical pipeline that S4D-TAM can adapt for efficient flow-conditioned occupancy prediction [7].

Wang et al. introduced OccSora, which uses a 4D scene tokenizer and a diffusion transformer for generative long-horizon 4D occupancy simulation conditioned on trajectories, offering S4D-TAM a generative world simulator approach for scenario generation and planning evaluation [10]. Liao et al. proposed I2-World with hierarchical intra/inter tokenizers and an encoder-decoder design for efficient compression and generation of temporally consistent 4D occupancy, presenting a tokenization strategy that S4D-TAM can apply for compact world representations and fast inference [14].

Zheng et al. presented GaussianAD, which represents scenes through semantic 3D Gaussians refined from images, predicts 3D flows for those Gaussians, and integrates forecasting with planning - this sparse semantic Gaussian approach is directly relevant to S4D-TAM's lightweight scene abstractions and prediction-based planning [17].

2.4. World Models and Planning

World models unify perception, prediction, and planning; for S4D-TAM, they provide a mechanism for simulating the future, evaluating trajectories, and incorporating action-conditioned forecasts. Chen et al. proposed DriveWorld - a Memory State-Space Model with dynamic memory and static scene propagation for spatiotemporal pre-training, demonstrating downstream gains in detection, mapping, tracking, forecasting, and planning - relevant to S4D-TAM pre-training strategies and multi-task world modeling [14], [18].

Zhao et al. used a contrastive world model objective to learn visual representations for drone navigation and visuomotor policy learning, illustrating how world model pre-training can improve navigation policies, which S4D-TAM can leverage for improved visual generalization [13]. Mei et al. proposed IR-WM, which focuses on predicting residual changes relative to a temporal BEV prior and couples forecasting with planning, offering S4D-TAM a compact residual prediction approach that reduces redundant static background modeling and improves planning integration [10].

Cai et al. presented NEUSIS - a compositional neuro-symbolic framework combining neuro-symbolic perception, a probabilistic world model with Bayesian filtering, and hierarchical search planning, exemplifying an interpretable and uncertainty-aware world modeling and planning workflow useful for S4D-TAM mission-level decision-making [11]. Liu et al. proposed HPP, which integrates occupancy-conditioned prediction with game-theoretic motion forecasting and planning, demonstrating methods for aligning prediction patterns with planning objectives that S4D-TAM can adopt for improved trajectory selection under uncertainty and interaction [11].

Feng et al. provided a comprehensive survey of world models for autonomous driving, covering 4D occupancy forecasting, generative simulation, and planning integration, providing a conceptual foundation and taxonomy that can guide S4D-TAM design choices in world model selection, pre-training, and evaluation [9].

2.5. UAV Navigation in GNSS-Denied Environments

Autonomous UAV navigation in GNSS-degraded environments requires multi-sensor fusion, robust SLAM algorithms, and uncertainty-aware planning strategies. Yue et al. proposed an air-ground cooperative UAV autonomous navigation system for GPS-denied environments that combines visual odometry, IMU, and communication with a ground robot for robust localization [19]. Radwan et al. presented UAV-assisted visual SLAM generating reconstructed 3D scene graphs in GPS-denied environments, demonstrating scene graph representations for scene understanding and planning [20].

Nieuwenhuisen et al. developed an autonomous navigation system for micro aerial vehicles in complex GNSS-denied environments, using LiDAR, IMU, and obstacle-aware planning [26], [53]. Yue et al. proposed semantically guided visual autonomous navigation for UAVs that employs semantic segmentation to identify navigable areas and obstacles for improved path planning [54]. Hu et al. provided a review of vision-based UAV localization and optimization-based path planning in GPS-denied environments, synthesizing SLAM, object detection, and trajectory planning methods [57].

Further relevant works include UAV navigation systems using multi-sensor fusion (IMU, camera, LiDAR, radar) [48], [50], [51], [52], [58], KAZE feature-based SLAM [49], map-aided navigation and dead reckoning [56], and platforms for search-and-rescue missions [55]. Castano presented autonomous UAV navigation in GPS-denied environments using LiDAR point clouds for localization and mapping [59].

3. Research Gap and Motivation

Contemporary solutions perform pose estimation, map construction, semantic segmentation, object recognition, occupancy prediction, trajectory planning, and risk assessment as separate, independent modules [9], [31], [40]. Systems such as LVI-SAM integrate LiDAR, camera, and IMU in a factor graph, while Kimera constructs metric-semantic 3D models. Neither, however, constitutes a complete, learned world model that exploits persistent map tokens for predicting unseen hazards and guiding sensor attention [12].

OpenScene enables assigning language-image embedding space features to point cloud points. Open3DSG extends such approaches with inter-object relationships, while Octree-Graph combines semantics, occupancy, and graph structure. These solutions, however, primarily address scene understanding rather than closed-loop UAV control operating under limited computational resources and partial observability [6], [7].

Research on 4D occupancy forecasting has demonstrated that predicting the future state of space can serve as an effective planning foundation [8], [10], [14], [17]. Nevertheless, automotive applications, dense voxel representations, and full sensor sequence processing still dominate. The proposed S4D-TAM system fills the gap between semantic SLAM, 4D world models, and risk-aware planning for UAVs in GNSS-degraded environments.

The key motivation is to transition from a classical geometric map to a predictive spatial memory of the UAV - one that not only records where objects are, but also predicts where they may appear, which areas are unobservable, and what risk they pose to flight safety. This approach can be termed a SLAM-native world model: a world model whose primary representation space is the SLAM map rather than the image stream [1].

4. S4D-TAM System Concept

4.1. Definition of the 4D Token

A 3D token would represent a spatial fragment or object at a given position:

$$t_i = [x_i, y_i, z_i]$$

It would not, however, capture environmental changes or observation history. In an autonomous system it is essential to distinguish a building, tree, or rock from a person, vehicle, or object that may change its position [1].

A 4D token should therefore incorporate the time dimension and additional attributes:

$$T_i(t) = [p_i, \Sigma_i, g_i, s_i, v_i, h_i, u_i, r_i]$$

where the individual elements denote:

1. $p_i \in \mathbb{R}^3$ - position in space
2. $\Sigma_i \in \mathbb{R}^3 \times 3$ - position covariance matrix (geometric uncertainty)
3. $g_i \in \mathbb{R}^{dg}$ - geometric descriptor (normals, curvature, local features)
4. $s_i \in \mathbb{R}^{ds}$ - semantic embedding (features from a neural network or open-vocabulary model)
5. $v_i \in \mathbb{R}^3$ - estimated velocity/motion vector
6. $h_i \in \mathbb{R}^{dh}$ - observation history (observation count, timestamps, sensor sources)
7. $u_i \in [0,1]$ - overall token uncertainty (scalar reliability measure)
8. $r_i \in [0,1]$ - local risk indicator (hazard probability)

The fourth dimension does not merely add time as an additional coordinate. The token also stores persistence, observation ordering, rate of change, and the predicted future state of the object [1], [17], [22], [23], [25].

4.2. System Architecture

The S4D-TAM system consists of the following layers (Figure 1):

Layer 1: Sensor Fusion Integration of data from RGB and thermal cameras, IMU, GNSS, altimeter, optionally LiDAR/radar, and reference maps. Each measurement receives a timestamp, coordinate frame transformation, and a reliability measure [1].

Layer 2: SLAM Front-End A lightweight visual front-end for feature tracking, depth estimation, characteristic object detection, and feature vector extraction. The geometric SLAM creates a local point cloud and estimates the UAV trajectory. The factor graph includes visual odometry, inertial, GNSS, altimeter, semantic loop-closure, and reference-map alignment factors [2], [3], [4].

Layer 3: Tokenization and Hierarchical Representation The full point cloud is divided into superpoints, object segments, adaptive octree cells, local surface patches, and dynamic instances. Each segment is encoded as one or more tokens. The hierarchical representation stores large, regular surfaces at low resolution while edges, obstacles, and dynamic objects are retained at higher resolution [5], [6], [7], [37], [38].

Layer 4: Spatiotemporal Transformer A hierarchical attention mechanism with local scope (tokens within a given spatial radius, predicted flight corridor, sensor visibility region) and global scope (landmarks, major terrain obstacles, dynamic objects, high-uncertainty areas, elevated-risk locations). The attention weight depends on token features and their mutual positions [8], [12], [14], [16].

Layer 5: Object Memory and Semantics Upon detecting a characteristic object, the system creates an instance token storing a class probability distribution, observation history, open-vocabulary embedding, and spatial relations with other objects [5], [11], [34].

Layer 6: Reference Map Alignment The system maintains two representations: the current map M_t built on the fly, and the reference map M_{ref} derived from elevation maps, previous flights, or a simplified terrain point cloud. Alignment uses a three-step procedure: global topological and terrain descriptor matching, landmark token matching, and local geometric optimization [4], [19], [20].

Layer 7: Occupancy and Risk Forecasting The model predicts future space occupancy and the probability of risk in partially observable areas. The system infers that beyond a building edge lies an unobservable region, a road or terrain trail may lead to an unseen object, a previously detected dynamic object may have moved into a neighboring zone, and terrain topography limits available escape trajectories [8], [10], [14], [17], [21].

Layer 8: Trajectory Planning The planner minimizes an objective function that accounts for trajectory length, energy cost, collision probability, predicted environmental risk, and the cost arising from map uncertainty. The system selects among: continuing flight, altitude change, heading change, temporary hover in a safe zone, return along the memorized route, limiting non-critical measurements, and increasing observation intensity in high-uncertainty areas [11], [19], [41].

Layer 9: Active Perception The active perception module selects camera orientation, processing frequency, and sensory modality based on expected uncertainty reduction relative to cost. The IR camera may be activated more frequently in low illumination or in areas where prior data

indicates the possible presence of a thermal object. The RGB camera need not be analyzed by the full network at every frame [1], [3], [6].

4.3. Current Reference Implementation and Target Research Modules

The S4D-TAM concept is accompanied by an executable Python research reference. The implementation is intentionally modular so that individual mechanisms can be tested, replaced, and ablated without changing the evaluator boundary.

Implemented reference components include persistent token state and lifecycle management (token.py, memory.py), token proposal and data association (proposal.py, association.py), modality-oriented encoder interfaces and masked fusion (encoders/), attention-related processing (attention.py), calibration and uncertainty utilities (calibration.py), reference-map and topological support (reference_map.py, topology.py), causal probabilistic occupancy and motion forecasting (forecasting.py), deterministic risk-, energy-, time-, progress-, and information-aware planning (planner.py), structured event telemetry (telemetry.py), and the integrated S4DTAMReference pipeline (pipeline.py).

The reference implementation favors causality, deterministic replay, explicit uncertainty, and auditability over maximum throughput. This is advantageous during early numerical validation because an observed difference can be traced to a defined mechanism rather than hidden implementation nondeterminism.

Several formulations in Sections 5.12-5.15 describe target research extensions rather than already completed backends. These include a learned hierarchical spatiotemporal model, a DBN/particle or equivalent multimodal probabilistic scene-state backend, an MPPI planner backend, adaptive octree/LOD storage, a closed-loop active-perception controller, and a safety-aware real-time scheduler. They are maintained as explicit implementation milestones and must preserve the same causal and evaluator-facing contracts when introduced.

5. Mathematical Description

5.1. Sensor Layer and Data Fusion

The system uses a sensor set $\mathcal{S} = \{s_1, s_2, \dots, s_n\}$, where each sensor s_i provides a measurement $z_i(t)$ at time t with uncertainty $\Sigma_i(t)$. Data fusion is performed in a factor graph:

$$F_{total} = FVO + FIMU + FGNSS + Falt + Fsem + Fref$$

where:

- FVO - visual odometry factor
- FIMU - inertial factor
- FGNSS - GNSS factor (with adaptive weight under degradation)
- Falt - altimeter factor
- Fsem - semantic loop-closure factor
- Fref - reference map alignment factor

Each factor takes the form:

$$F_i = \sum_k \|e_k\|_{\Sigma_i k}^2$$

where e_k is the residual error and $\Sigma_i k$ is the covariance matrix [2], [3], [4].

5.2. SLAM Front-End

The SLAM front-end estimates the UAV trajectory $X = \{x_1, x_2, \dots, x_T\}$ and the point map $M = \{m_1, m_2, \dots, m_N\}$ by minimizing the cost function:

$$X^*, M^* = \arg \min_{X, M} \sum_{i,j} \rho(\|z_{ij} - h(x_i, m_j)\|_{\Sigma_{ij}}^2)$$

where $h(x_i, m_j)$ is the projection function of point m_j onto pose x_i , and $\rho(\cdot)$ is a robust function (e.g., Huber) [2], [4].

The reconstruction scale is corrected using:

$$s^* = \arg \min_s \sum_k w_k \|s \cdot d_{kVO} - d_{kref}\|^2$$

where d_{kVO} is the distance from visual odometry, d_{kref} is the reference distance (from IMU, altimeter, GNSS, known object sizes, elevation map), and w_k are reliability weights [1], [4].

5.3. Point Cloud Tokenization

The point cloud M is divided into K tokens $\{T_1, T_2, \dots, T_K\}$ through hierarchical segmentation:

$$M = \bigcup_{i=1}^K \mathcal{P}_i$$

where $\mathcal{P}_i$ is the set of points assigned to token T_i . Segmentation may use:

1. Superpoints - grouping of points with similar geometry and semantics [37], [38]
2. Object segments - instance detection [5], [34]
3. Adaptive octree - hierarchical space partitioning [6], [7]
4. Local surface patches - planes, edges, corners [4]
5. Dynamic instances - moving objects [17], [22]

For each token T_i the following attributes are computed:

$$\begin{aligned} p_i &= (1/|\mathcal{P}_i|) \sum_{m \in \mathcal{P}_i} m \\ \Sigma_i &= (1/|\mathcal{P}_i|) \sum_{m \in \mathcal{P}_i} (m - p_i)(m - p_i)^T \\ g_i &= \text{GeomEncoder}(\mathcal{P}_i) \\ s_i &= \text{SemEncoder}(\mathcal{J}_i, \mathcal{P}_i) \end{aligned}$$

where $\mathcal{J}_i$ is the set of images observing points $\mathcal{P}_i$ [5], [16], [38].

5.4. Spatiotemporal Attention Mechanism

The hierarchical spatiotemporal transformer updates token representations through the attention mechanism:

$$T_i' = T_i + \text{Attention}(Q_i, K, V)$$

where:

$$Q_i = WQ T_i, \text{quad } K = WK T, \text{quad } V = WV T$$

Local attention covers tokens within radius r_{local} :

$$\mathcal{N}_{i\text{local}} = \{j : \|p_i - p_j\| < r_{\text{local}}\}$$

Global attention covers tokens of particular significance:

$$\mathcal{N}_{i\text{global}} = \{j : \text{landmark}(j) \vee \text{dynamic}(j) \vee u_j > \theta_u \vee r_j > \theta_r\}$$

The attention weight accounts for spatial, temporal, and uncertainty relations:

$$\alpha_{ij} = (\exp(\frac{Q_i^T K_j}{\sqrt{d_k}}) \cdot \varphi(p_i, p_j, t_i, t_j) \cdot \psi(u_i, u_j)) \sum_{k \in \mathcal{N}_i} \exp(\frac{Q_i^T K_k}{\sqrt{d_k}}) \cdot \varphi(p_i, p_k, t_i, t_k) \cdot \psi(u_i, u_k)$$

where $\phi(\cdot)$ encodes spatial and temporal relations, and $\psi(\cdot)$ limits the influence of uncertain observations [8], [12], [14], [16].

5.5. Object Memory and Semantics

Upon detecting a characteristic object, the system creates an instance token:

$$O_k = [p_k, \Sigma_k, b_k, s_k, P(c_k | z_{1:t}), h_k]$$

where b_k is the bounding box, and $P(c_k | z_{1:t})$ is the probability distribution over the object class c_k given the observation history $z_{1:t}$.

The class distribution is updated via Bayesian filtering:

$$P(c_k | z_{1:t}) = \frac{P(z_t | c_k) P(c_k | z_{1:t-1}) \sum_{c'} P(z_t | c') P(c' | z_{1:t-1})}{\sum_{c'} P(z_t | c') P(c' | z_{1:t-1})}$$

The open-vocabulary approach allows searching the map for objects and properties that were not constrained to a closed class set during training [5], [11], [34].

5.6. Reference Map Alignment

The system maintains two representations: the current map M_t and the reference map M_{ref} . Alignment uses a three-step procedure:

Step 1: Global Alignment Determining the transformation T_{global} by matching global descriptors:

$$T_{global} = \arg \min T \sum_i \| d_{it} - d_{\sigma(i)ref} \|^2$$

where d_{it} and $d_{\sigma(i)ref}$ are global descriptors (e.g., topological, terrain-based) [4], [19], [20].

Step 2: Landmark Matching Refining the transformation by matching landmark tokens:

$$T_{landmark} = \arg \min T \sum_{j \in \mathcal{L}} \| T p_{jt} - p_{\sigma(j)ref} \|^2$$

where $\mathcal{L}$ is the landmark set [2], [4].

Step 3: Local Geometric Optimization Final refinement via ICP or a semantic variant:

$$T_{final} = \arg \min T \sum_{i,j} w_{ij} \| T p_{it} - p_{jref} \|^2$$

where w_{ij} accounts for semantic consistency [4], [5].

The difference between the current map and the reference map indicates: a localization error, an environmental change, appearance of a new object, absence of a previously existing object, or unreliability of the current GNSS measurement [1], [19].

5.7. Occupancy and Risk Forecasting

The system predicts future space occupancy and risk probability in partially observable regions:

$$O(t+\Delta t) = \text{fpred}(T_{1:K}(t), a(t))$$

where $O(t+\Delta t)$ is the predicted occupancy at time $t+\Delta t$, $T_{1:K}(t)$ are tokens at time t , $a(t)$ is the planned action (trajectory), and fpred is the predictive model (e.g., temporal transformer, diffusion model) [8], [10], [14], [17].

The risk map is determined as:

$$R(p, t+\Delta t) = P(H_p, t+\Delta t | M_{1:t}, M_{ref}, C_t)$$

where $H_p, t+\Delta t$ denotes a hazardous event at position p at time $t+\Delta t$, and C_t is the motion and environment context [1], [21].

The system infers risk in unobservable areas via:

$$R_{occluded}(p) = \sum_{c \in \mathcal{C}} P(c | T_{visible}) \cdot P(H | c, \text{geometry}(p))$$

where $\mathcal{C}$ is the set of contexts (road, terrain trail, building edge), T_{visible} are the visible tokens, and $\text{geometry}(p)$ describes the terrain topography at position p [1], [11], [21].

5.8. Trajectory Planning

The planner minimizes the objective function:

$$J(\xi) = \int_0^T [L(\xi(t)) + E(\xi(t), \xi(t)) + \lambda_H PH(\xi(t)) + \lambda_R R(\xi(t), t) + \lambda_U U(\xi(t))] dt$$

where:

- $\xi(t)$ is the UAV trajectory
- $L(\xi(t))$ is the trajectory length
- $E(\xi(t), \xi(t))$ is the energy cost
- $PH(\xi(t))$ is the collision probability
- $R(\xi(t), t)$ is the predicted environmental risk
- $U(\xi(t))$ is the cost arising from map uncertainty
- $\lambda_H, \lambda_R, \lambda_U$ are weighting coefficients

Collision probability is determined as:

$$PH(\xi(t)) = 1 - \prod_i (1 - P(\text{collision} \mid \xi(t), T_i(t)))$$

The uncertainty cost accounts for missing information:

$$U(\xi(t)) = \sum_i \in \mathcal{N}(\xi(t)) u_i \cdot \exp(-(\|\xi(t) - p_i\|^2)/(2\sigma^2))$$

The system selects the action:

$$a^* = \arg \min_{a \in \mathcal{A}} J(\xi(a))$$

where $\mathcal{A}$ is the set of possible actions: continuing flight, altitude change, heading change, temporary hover in a safe zone, return along the memorized route, limiting non-critical measurements, and increasing observation intensity in high-uncertainty areas [1], [11], [19], [41].

5.9. Active Perception

The active perception module selects a sensory action a_{sense} that maximizes expected uncertainty reduction relative to cost:

$$a_{\text{sense}} = \arg \max_{a \in \mathcal{A}_{\text{sense}}} (I(T; z \mid a)) / (C(a))$$

where $I(T; z \mid a)$ is the mutual information between tokens T and measurement z given action a , and $C(a)$ is the energy and computational cost of the action [1], [3], [6].

Mutual information is approximated as:

$$I(T; z \mid a) \approx H(T) - E_z[a[H(T|z)]]$$

where $H(T)$ is the token entropy before measurement and $H(T|z)$ is the entropy after measurement.

The IR camera may be activated more frequently in low illumination or in regions where prior data indicate the possible presence of a thermal object. The RGB camera need not be processed by the full network at every frame [1].

5.10. Metrological Error Analysis of the S4D-TAM System

Metrological analysis forms the foundation for designing reliable autonomous navigation systems. The objective of this subsection is to formalize the error models of individual S4D-

TAM system layers, derive lower bounds on estimation uncertainty, and determine the cumulative positioning error budget of the UAV in a GNSS-degraded environment.

5.10.1. Sensor Layer Error Model

Each sensor $s_i \in \mathcal{S}$ is characterized by its own error model. Adopting the general measurement form:

$$z_i(t) = h_i(x(t)) + n_i(t) + b_i(t)$$

where $h_i(\cdot)$ is the observation function, $n_i(t) \sim \mathcal{N}(0, R_i)$ is the measurement noise with covariance matrix R_i , and $b_i(t)$ is the systematic error (bias).

IMU (gyroscope + accelerometer): The error model of the gyroscope and accelerometer in an IMU extended with bias is described as:

$$\omega(t) = \omega(t) + bg(t) + ng(t), \text{quad } ng \sim \mathcal{N}(0, \sigma g^2 I_3)$$

$$a(t) = a(t) + ba(t) + na(t), \text{quad } na \sim \mathcal{N}(0, \sigma a^2 I_3)$$

where biases evolve according to a random walk model:

$$bg = nb_g, \text{quad } nb_g \sim \mathcal{N}(0, \sigma b_g^2 I_3)$$

$$ba = nba, \text{quad } nba \sim \mathcal{N}(0, \sigma ba^2 I_3)$$

The position error resulting from double integration of IMU grows over time as:

$$\sigma_{IMU}(t) = (1)/(2)\sigma a t^2 + (1)/(6)\sigma ba t^3$$

GNSS under degradation conditions: In GNSS-degraded environments we introduce a signal quality indicator $qGNSS(t) \in [0, 1]$, where $qGNSS = 1$ denotes full constellation coverage and $qGNSS = 0$ denotes complete signal loss. The GNSS measurement covariance scales adaptively:

$$R_{GNSS}(t) = \text{frac} R_{GNSS}^{(0)} qGNSS(t) + \epsilon$$

where $R_{GNSS}^{(0)}$ is the baseline covariance at full signal and $\epsilon \ll 1$ is a regularization parameter preventing singularity. As $qGNSS \rightarrow 0$, the factor R_{GNSS} is effectively removed from the factor graph through the increase in its variance.

Visual odometry (VO) - relative drift: The visual odometry error accumulates quasi-randomly, proportional to the traveled distance d :

$$\sigma_{VO}(d) = \alpha_{VO} d + \beta_{VO} \sqrt{d}$$

where α_{VO} is the linear error coefficient (typically 0.5–2% per meter traveled), and β_{VO} describes the diffusive component associated with stochastic visual noise.

Barometric altimeter: The altitude measurement error is dominated by slow-varying temperature and ambient pressure drift:

$$z_{baro}(t) = z_{true}(t) + b_{baro} + n_{baro}(t)$$

where $b_{baro} \sim \mathcal{U}(-\delta b, +\delta b)$ is the bias with a uniform distribution over range $\delta b \approx 1\text{--}3$ m, and $n_{baro} \sim \mathcal{N}(0, \sigma_{baro}^2)$ with $\sigma_{baro} \approx 0.1\text{--}0.5$ m.

5.10.2. Error Propagation in the Factor Graph

Let $\xi = [p_{top}, v_{top}, \phi_{top}, b_{g_{top}}, b_{a_{top}}]_{top} \in \mathbb{R}^{15}$ be the navigation state vector (p - position, v - velocity, ϕ - Euler angles, b_g , b_a - IMU biases). The lower bound on the variance of an unbiased estimator ξ is given by the Cramér-Rao Lower Bound (CRLB):

$$\text{Cov}(\xi) \geq J^{-1}(\xi)$$

where $J(\xi)$ is the Fisher Information Matrix (FIM), the sum of contributions from all factor graph factors:

$$J(\xi) = \sum_i H_i^{top} R_i^{-1} H_i + J_{prior}$$

where $H_i = (\partial h_i)/(\partial \xi)$ is the Jacobian matrix of the observation function of the i -th factor, and J_{prior} represents the prior information contribution. The marginal position variance σ_p^2 is the diagonal element:

$$\sigma_p^2 \geq [J^{-1}(\xi)]_{pp}$$

Navigation state covariance propagation follows the Riccati equation:

$$P(t) = F(t) P(t) + P(t) F(t)^{top} + G(t) Q(t) G(t)^{top}$$

where $F(t)$ is the inertial system state transition matrix, $G(t)$ is the noise input matrix, and $Q(t) = \text{diag}(\sigma_a^2 I_3, \sigma_g^2 I_3, \sigma_b a^2 I_3, \sigma_b g^2 I_3)$ is the noise spectral density matrix.

When merging measurements from M sensors in the factor graph, the combined FIM grows additively, reducing the estimator covariance:

$$J_{fused} = J_{IMU} + J_{VO} + q_{GNSS} J_{GNSS} + J_{alt} + J_{sem} + J_{ref}$$

where each term is weighted according to the reliability of the corresponding source.

5.10.3. Quantization and Discretization Error of Tokens

The 3D space is divided into tokens of side length ℓ meters (§5.3). The quantization operation mapping the coordinates of point $p \in \mathbb{R}^3$ to the centroid of token c_k introduces a quantization error:

$$\varepsilon_q = p - c_k, \text{quad } \|\varepsilon_q\| \leq \frac{\ell}{\sqrt{3}}$$

For a regular cubic grid of side length ℓ , the quantization error variance along each axis is:

$$\sigma_{q,1D}^2 = (\ell^2)/12$$

and the total 3D variance:

$$\sigma_q^2 = (\ell^2)/4$$

An additional semantic error arises from imperfections of the open-vocabulary classifier. Defining the classifier confusion matrix $C \in \mathbb{R}^C \times C$ (C classes), the semantic error of token k is expressed by the conditional entropy of the class assignment:

$$\varepsilon_{sem}^{\wedge}(k) = -\sum_{c=1}^C p(c|T_k) \log p(c|T_k) = H(c|T_k)$$

A value of $\varepsilon_{sem}^{\wedge}(k) = 0$ indicates certain assignment, while $\varepsilon_{sem}^{\wedge}(k) = \log C$ indicates maximum semantic uncertainty. A token with high $\varepsilon_{sem}^{\wedge}(k)$ receives increased covariance in the F_{sem} factor graph factor.

5.10.4. Lower Bound on Localization Error

The total UAV positioning error is a combination of: visual odometry drift σ_{VO} , uncertainty arising from IMU integration σ_{IMU} , GNSS error σ_{GNSS} , and the contribution from semantic anchoring σ_{sem} . Under joint optimization in the factor graph, the CRLB on the position error takes the form:

$$\sigma_{p,min}^2(t) = [J_{fused}^{-1}(t)]_{pp} = (1)/(\sum_i [H_i^{top} R_i^{-1} H_i]_{pp})$$

In the scenario of complete GNSS absence ($q_{GNSS} = 0$), position error grows according to a modified drift model:

$$\sigma_p^2(t) = \sigma_{p,0}^2 + \alpha_{VO}^2 d(t)^2 + \gamma_{sem} N_{loop}^{-1}$$

where $\sigma_{p,0}^2$ is the initial variance (determined by GNSS before degradation), $d(t)$ is the total distance traveled, N_{loop} is the number of successful semantic loop closures, and γ_{sem} is the semantic correction coefficient. Each semantic loop closure reduces σ_p^2 proportionally to the strength of the semantic evidence, determined by the Fisher information of factor F_{sem} .

5.10.5. Occupancy Forecasting Error

The 4D occupancy predictor (§5.7) produces a probabilistic map $O(x, t+\Delta t)$ based on the token state at time t . Prediction error grows with the temporal horizon Δt . Modeling the evolution of token T_k as a linear-Gaussian stochastic process with transition matrix Φ and noise diffusion Q_{dyn} :

$$T_k(t+\Delta t) = \Phi(\Delta t) T_k(t) + \omega_k, \text{quad } \omega_k \sim \mathcal{N}(0, Q_{dyn}(\Delta t))$$

the prediction error covariance is:

$$\Sigma_{pred}(\Delta t) = \Phi(\Delta t) \Sigma_k(t) \Phi(\Delta t)^T + Q_{dyn}(\Delta t)$$

For static objects $\Phi \approx I$ and $Q_{dyn} \approx 0$, while for dynamic objects $Q_{dyn}(\Delta t) \propto \Delta t$. This results in a linear growth of prediction uncertainty with horizon:

$$\sigma_{pred}^2(\Delta t) = \sigma_{pred,0}^2 + \sigma_{dyn}^2 \cdot \Delta t$$

The binary occupancy classification error is defined by the Brier Score:

$$BS(\Delta t) = (1/|\mathcal{U}|) \sum_{x \in \mathcal{U}} (O(x, t+\Delta t) - O_{true}(x, t+\Delta t))^2$$

where $\mathcal{U}$ is the set of workspace voxels. The expected value of $BS(\Delta t)$ is lower bounded by the entropy of the dynamic environment.

5.10.6. Cumulative Error Budget

The table below summarizes the error sources of the S4D-TAM system with their typical values for the target configuration (lightweight UAV in an urban environment with partial GNSS degradation):

Error Source	Characteristic	Typical Value (1σ)	Time/Distance Dependence
IMU noise (accelerometer)	White noise σ_a	0.05 m/s ²	$\sigma \propto t^2$ (integration)
IMU bias drift	Random walk σ_{ba}	0.002 m/s ² /√s	$\sigma \propto t^{3/2}$
Visual odometry	Proportional drift α_{VO}	1% per meter	$\sigma \propto d$
GNSS (full signal)	$\sigma_{GNSS}^{(0)}$	1.5 m (horizontal)	Constant
GNSS (degraded)	$\sigma_{GNSS}(q)$	5–50 m	$\propto 1/q_{GNSS}$
Barometric altimeter	$\sigma_{baro} + \text{bias}$	0.3 m + 2 m	Thermal drift
Token quantization	σ_q	$\ell/2$ (e.g., 0.5 m for $\ell = 1$ m)	Constant (depends on ℓ)
Semantic error	Classifier entropy $H(\mathbf{c}_T)$		0.1–0.8 bit
Occupancy prediction ($\Delta t = 1$ s)	$BS(\Delta t)$	0.05–0.15	$\propto \Delta t$
Total position error (IMU+VO only, $d = 100$ m)	CRLB σ_p	≈ 2 –5 m	$\propto d$
Total position error (IMU+VO+sem, $N_{loop} = 5$)	CRLB σ_p	≈ 0.5 –1.5 m	Bounded by loop closures

The error budget indicates that the key mechanism for limiting position drift under GNSS-degraded conditions is the effectiveness of semantic loop closures. Each loop closure with high semantic confidence (low $\varepsilon_{\text{sem}}^{\wedge}(k)$) reduces position variance proportionally to the Fisher information of factor F_{sem} , enabling maintenance of the error below the operational threshold $\sigma_p < 2$ m even when flying through zones of complete GNSS loss extending to several kilometers.

5.11. Extended Semantic Token Model

The basic geometric token introduced in §5.3 is replaced by an extended semantic-temporal token $\mathcal{T}_k$, constituting a complete profile of an object recognized in the UAV's environment. Each token stores not only the object's position, but a complete record of the multidimensional observation history:

$$k = \{ \mu_k, \Sigma_k, pc^{\wedge}(k), \delta_k, v_k, \Omega_k, \rho_k \}$$

where the individual components have the following meaning:

- $\mu_k \in \mathbb{R}^3$ - estimated position (centroid) in the global frame
- $\Sigma_k \in \mathbb{R}^3 \times 3$ - position covariance matrix (localization uncertainty)
- $pc^{\wedge}(k) = (c_j | \mathcal{T}_k)_{j=1}^C$ - semantic class probability distribution
- $\mathcal{O}_k = \{(t_i, s_i, w_i, z_i)\}_{i=1}^{n_k}$ - observation history: time, sensor ID, reliability weight, measurement
- $\delta_k \in \{, \text{dynamic}, \text{periodic}\}$ - motion state classification
- $v_k \in \mathbb{R}^3$ - estimated velocity vector (for $\delta_k \neq \text{static}$)
- $r_k = \{r_{kl}\}$ - set of spatial-semantic relations to neighboring tokens
- Ω_k - occlusion region (areas hidden by object k)
- $\rho_k \in [0,1]$ - significance coefficient for flight safety and planning

Weighted multi-sensor fusion:

Each observation in the history $\mathcal{O}_k$ carries a reliability weight depending on the sensor source, detection confidence, and time elapsed since the last observation:

$$w_i = w_{\text{base}}(s_i) \cdot \text{conf}_i \cdot f_{\text{decay}}(t - t_i)$$

where $w_{\text{base}}(s_i)$ is the baseline reliability of sensor s_i (calibrated a priori), $\text{conf}_i \in [0,1]$ is the detection confidence, and f_{decay} is the temporal decay function:

$$f_{\text{decay}}(\Delta t) = \exp(-\lambda_s \cdot \Delta t)$$

The constant λ_s is a sensor-specific parameter: RGB cameras decay faster ($\lambda_{\text{RGB}} > \lambda_{\text{LiDAR}}$) due to sensitivity to illumination changes. The position estimate and its covariance are determined by weighted information fusion:

$$\Sigma_k^{-1} = \sum_{i=1}^{n_k} w_i \cdot \Sigma_i^{-1}, \quad \mu_k = \Sigma_k \sum_{i=1}^{n_k} w_i \cdot \Sigma_i^{-1} z_i^{\wedge}(k)$$

Bayesian update of the semantic class:

At each new observation z_{new} , the token class is updated according to Bayes' rule:

$$p(c_j | \mathcal{T}_k, z_{\text{new}}) \propto p(z_{\text{new}} | c_j) \cdot p(c_j | \mathcal{T}_k)$$

The prior $p(c_j | \mathcal{T}_k)$ comes from the previous iteration, and the likelihood $p(z_{\text{new}} | c_j)$ is the output of the open-vocabulary classifier. The accumulation of observations causes semantic certainty to increase monotonically with n_k .

Motion state classification:

Based on the observation history $\bar{O}_k$, the system classifies the token into one of three dynamic classes. Define the sequence of estimated displacements:

$$\Delta \mu_k(i) = \mu_k(t_i) - \mu_k(t_{i-1})$$

The classification criterion is based on a statistical test:

$$\delta_k = \begin{cases} \text{static} & \text{if } \|\Delta \mu_k\| < \varepsilon_s \text{ and } \sigma_{\Delta} < \varepsilon_{\sigma} \\ \text{periodic} & \text{if } FFT\|\Delta \mu_k(i)\| \text{ contains a dominant frequency } f^* \\ \text{dynamic} & \text{otherwise} \end{cases}$$

where ε_s and ε_{σ} are experimentally calibrated thresholds.

Spatial relations between tokens:

A pair of tokens $(\mathcal{T}_k, \mathcal{T}_l)$ is connected by a relation r_{kl} describing their mutual geometric and semantic conditioning:

$$r_{kl} = \{ d_{kl}, e_{kl}, \psi_{kl}, \alpha_{kl} \}$$

where $d_{kl} = \|\mu_k - \mu_l\|$ is the distance, e_{kl} is the unit direction vector, ψ_{kl} is the ontological relation (e.g., above, adjacent, blocks), and α_{kl} is the position uncertainty correlation coefficient. These relations form a scene graph $\mathcal{G} = (\mathcal{U}, \mathcal{E})$, in which vertices are tokens and edges carry relations r_{kl} .

Modeling of occluded areas:

For each token $\mathcal{T}_k$ and the current UAV position x_{UAV} , the system determines the occlusion cone $\Omega_k(x_{UAV})$ as the set of points geometrically hidden by the object:

$$\Omega_k(x_{UAV}) = \{ x \in \mathbb{R}^3 : \exists \lambda > 1, x = x_{UAV} + \lambda (\mu_k - x_{UAV}), \|x - axis_k\| < r_k \}$$

where r_k is the estimated object radius. These regions enrich the uncertainty map with knowledge about the unknown surroundings.

Flight safety significance coefficient:

Token significance ρ_k is a function of the semantic class, observation history, and operational context:

$$\rho_k = p_{class}(c_k^*) \cdot (1 - \exp(-\beta_{\rho} \cdot n_k)) \cdot f_{context}(\mathcal{T}_k, x_{UAV})$$

where $p_{class}(c_k^*)$ is the baseline class significance (e.g., $p_{wire} = 0.95$, $p_{tree} = 0.6$, $p_{building} = 0.8$), the factor $(1 - \exp(-\beta_{\rho} \cdot n_k))$ reflects growing confidence with the number of observations, and $f_{context}$ accounts for distance and current flight conditions.

5.12. Bayesian Inference about Future Scene States

Implementation status. The current executable reference uses a strictly causal probabilistic occupancy and motion forecaster driven only by observations available up to the prediction time. It produces Bernoulli occupancy probabilities, flow means, uncertainty estimates, observability masks, and physically timed forecast horizons, including irregular or dropped samples. This deterministic causal baseline is preferred for initial validation because future-frame leakage is structurally impossible. The DBN and particle approximation below define the target probabilistic extension for multimodal future-state uncertainty and are not claimed as the current default backend.

Predicting future scene states is a key element of proactive UAV navigation. The system models the scene as a Dynamic Bayesian Network (DBN), in which the global scene state at time t is defined as:

$$S_t = \mathcal{T}_{k(t)k=1K}$$

Scene state transition model:

Assuming conditional independence of tokens with respect to the common environmental background, the transition model factorizes as:

$$p(St+\Delta t | St) = \prod_{k=1}^K p(\mathcal{T}_k(t+\Delta t) | \mathcal{T}_k(t), \delta_k, \mathcal{L}_t)$$

Separate transition models are applied for static, dynamic, and periodic tokens:

$$p(\mu_k(t+\Delta t) | \mu_k(t), \delta_k) = \begin{cases} \text{begin cases } \mathcal{N}(\mu_k, \Sigma_k + Q_s \Delta t) & \delta_k = \text{static} \\ \mathcal{N}(\mu_k + v_k \Delta t, \Sigma_k + Q_d \Delta t) & \delta_k = \text{dynamic} \\ \sum_j \alpha_j \mathcal{N}(\mu_k(j), \Sigma_k(j)) & \delta_k = \text{periodic} \end{cases}$$

where Q_s and Q_d are process noise matrices for static and dynamic objects, and the periodic model is a Gaussian mixture with components $\mu_k(j)$ corresponding to typical positions within the cycle.

Particle approximation of the scene state distribution:

Due to the nonlinearity of models and multimodality of distributions, the posterior probability of the scene state is approximated by a set of N_p particles $\{st^{(i)}, wt^{(i)}\}_{i=1}^{N_p}$:

$$p(St | z_{1:t}) \approx \sum_{i=1}^{N_p} wt^{(i)} \delta(St - st^{(i)})$$

The particle filter recursion proceeds in three steps:

Prediction:

$$st_{t+1}^{(i)} \sim p(St_{t+1} | st^{(i)})$$

Weight update:

$$wt_{t+1}^{(i)} \propto wt^{(i)} \cdot p(z_{t+1} | st_{t+1}^{(i)})$$

Normalization and resampling:

$$wt_{t+1}^{(i)} \leftarrow \frac{wt_{t+1}^{(i)}}{\sum_j wt_{t+1}^{(j)}}, \text{quad resample when } Neff = (1)/(\sum_i (wt^{(i)}))^2 < N_p / 2$$

The predictive occupancy map at horizon Δt is the particle-weighted average:

$$O(x, t+\Delta t) = \sum_{i=1}^{N_p} wt^{(i)} O(x | st+\Delta t^{(i)})$$

Inference about occluded areas:

For the occlusion region Ω_k determined in §5.11, the occupancy probability is inferred from the semantic context of the occluding token and spatial environment structure statistics:

$$p(\text{occupied}(x) | \mathcal{T}_k) = p_0(c_k^*) + (1 - p_0(c_k^*)) \exp(-(d(x, \partial\Omega_k)^2)/(2\sigma_\Omega^2)), \text{quad } x \in \Omega_k$$

where $p_0(c_k)$ is the baseline probability of an obstacle existing behind an object of class c_k (e.g., $p_0(\text{building}) = 0.8$, $p_0(\text{tree}) = 0.3$), and σ_Ω controls the uncertainty range. This inference allows the planner to account for hazards invisible to sensors.

Anomaly and event detection:

The deviation of the observed state St_{obs} from the predicted state St is measured by the Mahalanobis distance:

$$DM(t) = \sqrt{((St_{obs} - St)^T \Sigma^{-1} (St_{obs} - St))}$$

Exceeding the threshold $DM(t) > D_{alarm}$ triggers heightened attention mode, in which the system increases processing frequency and directs sensors toward the anomaly area.

5.13. Bayesian Risk Estimation and Trajectory Planning

Implementation status. The executable reference planner currently uses deterministic finite-horizon beam search over admissible acceleration controls with explicit kinematic limits and decomposed collision-risk, energy, time, goal-progress, and information-value terms. This approach is more auditable than stochastic sampling for early confirmatory experiments because identical inputs generate the same candidate evaluation and decision trace. MPPI

remains a planned backend for richer dynamics and will be introduced behind the same planner contract only after direct comparison with the deterministic reference.

The S4D-TAM system integrates the risk map directly into the trajectory planning process, treating collision risk as a probabilistic field determined in real time from the current Bayesian state.

Risk field definition:

The risk field $\mathcal{R}(x, t)$ at spatial point x at time t is the sum of weighted contributions from all tokens, including occluded areas:

$$\mathcal{R}(x, t) = \sum_k \rho_k \cdot p_{occ}(x, \mathcal{T}_k, t) \cdot p_{coll}(x, \mathcal{T}_k)$$

where the occupancy probability p_{occ} comes from the Bayesian prediction (§5.12), and the geometric collision probability is modeled as the complementary CDF of the safety distance:

$$p_{coll}(x, \mathcal{T}_k) = 1 - \Phi(\text{fracd}(x, \mu_k) - r_{safe} / (\sigma_{k,r}^2 + \sigma_{UAV}^2))$$

where r_{safe} is the minimum safety radius, $\sigma_{k,r}$ is the object size uncertainty, and σ_{UAV} accounts for UAV position uncertainty.

Risk in occluded areas is elevated proportionally to the uncertainty of the occluded region:

$$\mathcal{R}(x, t) \leftarrow \mathcal{R}(x, t) + \kappa_{\Omega} \cdot 1_{x \in \bigcup_k \Omega_k} \cdot \max_k p_o(c_k^*)$$

Trajectory cost function:

The optimal trajectory τ^* minimizes a multi-component cost function:

$$J(\tau) = w_L L(\tau) + w_E E(\tau) + w_R \int_{\tau} \mathcal{R}(x(t), t) dt + w_U \int_{\tau} \mathcal{U}(x(t), t) dt - w_I \mathcal{J}(\tau)$$

where:

- $L(\tau)$ - trajectory length
- $E(\tau)$ - energy cost (dependent on UAV dynamics)
- $\mathcal{R}$ - integrated collision risk along the route
- $\mathcal{U}(x, t) = \sigma_p^2(x, t) + \kappa_{\Omega} \cdot 1_{x \in \Omega}$ - world model uncertainty
- $\mathcal{J}(\tau)$ - information gain (rewarding active exploration of high-uncertainty zones)

Information gain is expressed through the expected scene entropy reduction:

$$\mathcal{J}(\tau) = H(S_t) - E_{\tau}[H(S_t | z_{1:\tau})]$$

Bayesian trajectory optimization (BTO):

$$\tau^* = \arg \min_{\tau \in \mathcal{F}} ES[J(\tau | S)]$$

For a future high-dimensional stochastic planning backend, direct minimization of the expected value may be approximated by particle trajectory sampling via MPPI (Model Predictive Path Integral). The following formulation defines that target extension; it is not required by the current deterministic reference planner:

$$\tau^* \approx \text{frac} \sum_{m=1}^M \exp(-(1)/(\lambda \text{MPPI}) J(\tau_m)) / \sum_{m=1}^M \exp(-(1)/(\lambda \text{MPPI}) J(\tau_m))$$

where M sampled trajectories τ_m are generated randomly from a normal distribution around the nominal trajectory, and λMPPI is a temperature parameter controlling the exploration-exploitation tradeoff.

Multi-level decision system:

The risk field $\mathcal{R}(x, t)$ defines four decision levels:

$$UAV Mode = \begin{cases} \text{nominal} & \text{if } (x_{UAV}, t) < \text{safe} \parallel \text{alert} \text{ \& } \text{safe} \leq \text{alert} \parallel \text{avoidance} \\ \text{avoidance} & \text{if } \text{alert} \leq \text{avoid} \parallel \text{emergency} \text{ \& } \text{avoid} \leq \text{emergency} \end{cases}$$

In alert mode, the planner selects avoidance trajectories with wider margins; in avoidance mode, an immediate maneuver is required; in emergency mode, the UAV initiates a safe landing or return procedure.

5.14. Adaptive World Model with Hierarchical Resolution

Implementation status. The current token memory already enforces bounded history and configurable limits on token count, map memory, and update time, together with lifecycle activation, sleeping, reactivation, merging, and removal rules. These mechanisms implement the resource-bounded principle of this section. The full adaptive octree and Level-of-Detail storage policy described below remains a planned world-model backend and will be validated separately before becoming the default representation.

Uniform tokenization resolution is a fundamental bottleneck in systems operating under limited computational resources. The S4D-TAM system introduces an Adaptive World Occupancy Model (AWOM) based on a hierarchical octree structure.

Hierarchical octree structure:

The UAV's operational space is represented by an adaptive octree $\mathcal{O}$ with node resolutions $\text{ell} \in \{\text{ell}_1, \text{ell}_2, \dots, \text{ell}_L\}$, where $\text{ell}_1 < \text{ell}_2 < \dots < \text{ell}_L$:

$$\text{ell}_1 = 0.1 \text{ m}, \text{quad } \text{ell}_2 = 0.5 \text{ m}, \text{quad } \text{ell}_3 = 2 \text{ m}, \text{quad } \text{ell}_4 = 10 \text{ m}$$

Each octree node at level ell_j is compressed into a Level of Detail (LOD) token representation at level j :

$$LOD_k \in \{0, 1, 2, 3\}$$

- LOD 0: token stores only (μ_k, ρ_k, c_k^*) - centroid, significance, and dominant class
- LOD 1: LOD 0 + approximate point cloud (up to 64 pts) + $pc^*(k)$
- LOD 2: LOD 1 + full attention embedding + observation history $\mathcal{O}_k$
- LOD 3: LOD 2 + motion prediction model + spatial relations $\mathcal{R}_k$ + region Ω_k

Resolution adaptation criterion:

The optimal token resolution ell_k is determined as a trade-off between perception quality, computational resources, and location significance:

$$\text{ell}_k = \text{ell}_1 \cdot 2^{\wedge L_k}, \text{quad } L_k = \left\lfloor \frac{L_{\max}}{\log_2} \cdot \text{fdist}(d_k) \cdot \text{frisk}(\rho_k)^{-1} \cdot \text{fcomp}(\text{Cavail})^{-1} \right\rfloor$$

where:

$$\text{fdist}(d_k) = \min(1, \text{fracd}(\mu_k, x_{UAV})d_{\max}), \text{quad } \text{frisk}(\rho_k) = \rho_k + \varepsilon_{\rho}, \text{quad } \text{fcomp}(\text{Cavail}) = \frac{\text{Cavail}}{\text{Cnominal}}$$

Interpretation: tokens close to the UAV, with high safety significance, and under conditions of good resource availability receive higher resolution; distant, low-significance tokens, or those during CPU overload, receive lower resolution.

Dynamic processing priorities:

Each token $\mathcal{T}_k$ has a priority scalar π_k computed at each planning cycle:

$$\pi_k = \alpha_{\pi} \cdot \rho_k + \beta_{\pi} \cdot (1)/(\lambda \min)(\Sigma_k) + \varepsilon + \gamma_{\pi} \cdot \text{fage}(t - t_{\text{last},k}) + \delta_{\pi} \cdot 1_{\delta_k \neq \text{static}}$$

where:

- ρ_k - safety significance (§5.11)

- $\lambda_{\min}(\Sigma_k)$ - smallest eigenvalue of the covariance matrix (lower value = high confidence, lower update priority)
- $f_{\text{age}}(t - t_{\text{last}}, k) = 1 - \exp(-v \cdot (t - t_{\text{last}}, k))$ - aging factor encouraging refresh of stale tokens
- $1_{\delta_k} \neq \text{static}$ - priority bonus for dynamic and periodic tokens

Computational budget allocation across layers:

The total computational budget C_{total} is distributed among the processing pipeline layers:

$$C_{\text{total}} = C_{\text{sensor}} + C_{\text{SLAM}} + C_{\text{token}} + C_{\text{attn}} + C_{\text{pred}} + C_{\text{plan}}$$

Constraints: $\sum_i C_i \leq C_{\text{total}}$ and $C_{\text{sensor}} \geq C_{\text{sensormin}}$ (minimum resources for perception safety). When $C_{\text{avail}} < C_{\text{nominal}}$, reduction occurs in cascade from the layer with the lowest safety priority:

$$C_i \leftarrow \max(C_{\text{min}}, C_i \cdot \frac{C_{\text{avail}}}{C_{\text{nominal}}}), \text{quad } i \in \{pred, plan, token, SLAM, sensor\}$$

5.15. Computational Resource Management and Bottleneck Elimination

Implementation status. Structured telemetry and resource budgets already expose update-time, token-count, memory, lifecycle, and decision events needed for profiling and replay. Predictive queue-overflow control, EDF-RT scheduling, guaranteed safety-critical CPU reservations, and the full survival-mode controller described below remain planned real-time systems extensions rather than features of the current Python reference.

Intelligent resource management is a necessary condition for realizing all described S4D-TAM layers on embedded-class UAV hardware. This section formalizes the data flow model, bottleneck identification, and elimination mechanisms.

Processing pipeline throughput analysis:

The S4D-TAM pipeline consists of N sequential stages with processing times T_i and queue sizes $q_i(t)$. System throughput is limited by the bottleneck stage:

$$\Theta_{\text{sys}} = (1)/(\max_i T_i), \text{quad } i^* = \arg \max_i T_i$$

For the S4D-TAM pipeline with stages: {acquisition, SLAM front-end, tokenization, attention, Bayesian prediction, planner}, the nominal cycle time is:

$$T_{\text{cycle}} = \sum_{i=1}^6 T_i + T_{\text{comm}}$$

where T_{comm} accounts for data bus latencies between modules.

Bottleneck prediction:

The system monitors queue lengths $q_i(t)$ and predicts overload via Bayesian methods. Modeling task arrival as a Poisson process with intensity $\lambda_i(t)$, the probability of queue overflow within horizon Δt is:

$$P(q_i(t + \Delta t) > q_{\text{max}}) = 1 - \sum_{n=0}^{q_{\text{max}} - q_i(t)} \frac{(\lambda_i \cdot \Delta t)^n}{n!} e^{-\lambda_i \Delta t}$$

When $P > P_{\text{alarm}}$, the system proactively reduces the load of stage i by:

- Reducing tokenization resolution (tokenization stage)
- Skipping updates of low-priority tokens π_k (attention stage)
- Shortening the prediction horizon Δt_{pred} (prediction stage)
- Simplifying the trajectory sampler (reducing M in MPPI, planner stage)

Adaptive processing frequency:

The effective processing frequency is dynamically scaled to available resources:

$$f_{proc}(t) = \min(f_{nom}, \text{fracCavail}(t)C_{per})$$

When $f_{proc} < f_{min}$, the system automatically simplifies the world model: it switches to survival mode with only a local attention map within radius $r_{survival}$ and a reactive planner based on VFH+ (Vector Field Histogram+).

Token memory management - TTL and LOD-compression policy:

Total token memory is bounded by the budget:

$$M_{tokens} = \sum_{k=1}^K m(LOD_k) \leq M_{budget}$$

where $m(LOD_j)$ is the token size at level j (e.g., $m(0) = 64$ B, $m(1) = 512$ B, $m(2) = 4$ KB, $m(3) = 32$ KB). The memory management policy combines three mechanisms:

TTL (Time-To-Live): a token is deactivated if $t - t_{last,k} > TTTL$ and $\rho_k < p_{prune}$.

Merging: static tokens at distance $d_{kl} < \ell_{kl}$ with KL-divergence $DKL(p_c^{(k)} \parallel p_c^{(l)}) < \epsilon_{KL}$ are merged into a common token:

$$\mu_{kl} = (n_k \mu_k + n_l \mu_l) / (n_k + n_l), \quad q_{quad} \Sigma_{kl}^{-1} = \Sigma_k^{-1} + \Sigma_l^{-1}$$

LOD-compression: tokens beyond the active processing range are degraded to a lower LOD, freeing operational memory.

Task scheduling with safety guarantee:

The task scheduler allocates CPU time to S4D-TAM system modules according to an EDF-RT (Earliest Deadline First - Real-Time) policy extended with a safety coefficient:

$$Priority_i(t) = \text{frac}_{p_i} \max T_{deadline,i}(t) - t + \kappa_{safety} \cdot 1_i \in \mathcal{C}_{safety}$$

where $\rho_{i\max} = \max_k \rho_k$ in the task queue of stage i , $T_{deadline,i}(t)$ is the absolute completion deadline, and $\mathcal{C}_{safety}$ is the set of safety-critical stages (sensor acquisition, SLAM front-end, collision detection). Safety-critical tasks $\mathcal{C}_{safety}$ have a guaranteed minimum CPU time regardless of system load, ensuring minimally safe operation even under full resource saturation.

6. Research Hypotheses

The confirmatory mechanism study uses seven preregistered hypotheses. Each hypothesis compares the full S4D-TAM model with a variant that disables exactly one mechanism while data, seeds, preprocessing, optimization budget, and all other settings remain fixed. External systems are deliberately excluded from this causal ablation family.

H1: Semantics improve navigation in scenarios containing dynamic objects. The primary endpoint is mission success; collisions per kilometer and semantic mIoU are secondary outcomes.

H2: Temporal state improves 1 s future-state prediction. The primary endpoint is forecast occupancy IoU at 1 s; flow EPE and temporal flip rate are secondary outcomes.

H3: Calibrated uncertainty reduces navigation risk. The primary endpoint is collisions per kilometer; ECE, NLL, and near misses are secondary outcomes.

H4: Topological support increases mission success. Path efficiency and ATE RMSE are secondary outcomes.

H5: Reference-map support reduces localization error. The primary endpoint is ATE RMSE; RPE and final drift percentage are secondary outcomes.

H6: Predictive risk modeling reduces collisions. The primary endpoint is collisions per kilometer; near misses and minimum clearance are secondary outcomes.

H7: Token lifecycle management reduces computational cost without materially degrading navigation quality. Confirmation requires both lower p95 latency and non-inferior mission success with a preregistered margin of 2 percentage points; map bytes and peak RSS are secondary outcomes.

H1-H6 are superiority hypotheses in the preregistered direction. H7 combines efficiency superiority with mission-success non-inferiority. The seven confirmatory decisions form one family and use the Holm procedure defined in Section 7.4.

7. Proposed Experimental Validation

7.1. Datasets

The validation framework targets TartanAir, Blackbird, MARSIM, and AeroVerse, each converted to the common SequenceData contract with immutable source version, license record, coordinate convention, calibration metadata, and SHA-256 manifest. The cohort identifier frozen-2026-01 denotes the study cohort rather than a vendor release and must resolve to exact source versions and file hashes.

Data are split at the environment or flight level, never by adjacent frames. The train split is the only source of learned weight updates; calibration is reserved for thresholds, normalization, early stopping, and uncertainty calibration; test remains inaccessible until code, artifacts, and analysis are frozen.

The preregistered offline mechanism study uses 20 preselected test sequences per dataset, yielding 80 paired sequence-level units. Systems validation additionally defines 30 paired SIL scenarios with five seeds, 20 HIL scenarios with five seeds, and 20 paired controlled real-flight missions per variant when the safety gate remains open.

A dedicated UAV dataset remains desirable for synchronized RGB, thermal, IMU, GNSS, point cloud, elevation/reference map, static and dynamic labels, visibility or observability masks, and controlled sensor-degradation scenarios. Metrics that require unavailable ground truth are reported as unavailable rather than silently omitted or imputed.

7.2. Two-Level Comparison Design

The evaluation is split into two non-interchangeable levels. Both levels use common dataset and metric contracts where applicable, but they answer different scientific questions and produce separate statistical reports.

7.2.1. External System Comparison

The external comparison asks whether the complete S4D-TAM system is competitive with independently implemented navigation and SLAM systems under the same data, hardware policy, sensor availability, and evaluator definitions.

The mandatory primary baseline set comprises ORB-SLAM3, VINS-Mono, FAST-LIO2, and LIO-SAM. Each baseline is pinned to an upstream revision and fixed environment or container, calibration, sensor-topic mapping, loop-closure policy, warm-up policy, and

hardware configuration. Results enter the benchmark only through the common AlgorithmResult artifact schema.

Additional external systems may be included only when their sensing assumptions are compatible, their exact implementation can be reproduced, and their output can be normalized without method-specific post-processing. Such additions must be frozen before the external study is executed.

This level measures complete-system competitiveness. A difference relative to an external baseline cannot be interpreted as the causal effect of an individual S4D-TAM component.

7.2.2. Internal Mechanism Study

The internal comparison tests mechanism contribution using the full system and seven single-component ablations: H1_no_semantics, H2_no_temporal_state, H3_no_calibrated_uncertainty, H4_no_topology, H5_no_reference_map, H6_no_risk_prediction, and H7_no_token_lifecycle.

Each variant differs from full by exactly one switch and uses its own frozen trained artifact. External systems never appear in this matrix. This separation prevents confounding complete-system identity with the causal effect of removing one S4D-TAM mechanism.

7.3. Metrics

Primary endpoints are deliberately compact and preregistered: localization uses ATE RMSE after SE(3) alignment without scale; forecasting uses occupancy IoU at 1 s and, for broader characterization, 3 s; safety uses mission success and collisions per kilometer; efficiency uses p95 latency and persistent map-memory footprint.

Localization and SLAM secondary metrics include RPE translation RMSE, final drift in meters and percentage of reference path length, and rotation RPE when orientation ground truth is available. Relocalization success rate, time to relocalize, and post-relocalization pose error are reported when the dataset supports those events.

Semantic metrics include mIoU and macro F1. Forecasting metrics include per-horizon occupancy IoU/F1, flow EPE, Brier score, NLL, and ECE. Pose uncertainty is evaluated by NEES, 95% coverage, and NLL where covariance output is available. Risk prediction may additionally report AUROC, false-alarm rate, and miss rate.

Navigation metrics include mission success, collisions per kilometer, near misses, minimum clearance, and path efficiency. Efficiency metrics include latency p95, map bytes, peak RSS, CPU time, and implementation-specific telemetry such as processed-token counts when consistently available.

Scale error, embedding quality, spatial-relation accuracy, emergency replanning counts, FPS, and energy consumption may be reported as secondary or exploratory metrics when their ground truth and measurement procedure are explicitly defined. They do not become confirmatory endpoints merely because an implementation emits them.

A missing metric caused by absent ground truth or absent algorithm output is recorded as unavailable. Algorithm failure, timeout, abort, or collision is not treated as a missing observation and remains visible in the safety analysis.

7.4. Confirmatory Protocol and Statistical Analysis

The preregistered H1-H7 study uses sequence or mission level inference rather than treating individual frames as independent samples. Continuous outcomes are summarized with paired mean differences and 95% BCa bootstrap confidence intervals using 10,000 resamples. Mission success is modeled as a paired scenario-level probability outcome, while distance-normalized event counts use Poisson or negative-binomial modeling with log distance as an offset according to the preregistered overdispersion rule.

The seven H1-H7 confirmatory decisions form one family and are corrected with the sequential Holm procedure at family-wise $\alpha = 0.05$. H1-H6 require the corrected result and confidence interval to support the predicted superiority direction. H7 additionally requires mission-success non-inferiority within 2 percentage points while improving p95 latency.

No missing technical result is imputed. Aborts and algorithm failures remain in the denominator for safety outcomes; complete-case reporting is accompanied by the preregistered worst-case sensitivity analysis. Exclusions are limited to integrity failures, excessive missing timestamps, independently confirmed sensor faults, or operational constraints defined before result inspection.

The external system comparison is reported separately from this H1-H7 inferential family. It uses the same frozen datasets and common evaluators, but its effect estimates describe differences between complete systems rather than component-level causal effects.

8. Discussion

The proposed S4D-TAM system represents an attempt to transition from a classical geometric map to a predictive spatial memory of the UAV. The key advantage is amortized perception - the cost of image analysis is incurred once, while the result remains available in spatial memory across many subsequent planning cycles [1]. This approach may substantially reduce computational redundancy in systems where the same images are repeatedly analyzed by different modules [12], [40].

The hierarchical token representation allows storing large, regular surfaces at low resolution, while edges, obstacles, and dynamic objects are retained at higher resolution [5], [6], [7], [37], [38]. This is consistent with the direction set by Superpoint Transformer, Octree-Graph, and other architectures demonstrating that aggregating points into higher-level structures can substantially reduce representation complexity.

The spatiotemporal attention mechanism enables modeling of long-range relations between tokens while maintaining computational efficiency by restricting attention to local neighborhoods and global landmarks [8], [12], [14], [16]. Token-uncertainty-weighted attention limits the influence of uncertain observations on world model updates, which may improve system stability under conditions of partial observability [1], [11].

Occupancy and risk forecasting in partially observable areas constitutes the most interesting element of the system - the transition from detection to prediction [1], [8], [10], [14], [17], [21]. The system does not claim to have recognized an object it has not seen, but determines the probability of risk existing in partially observable space based on geometric context, semantics, and observation history. This approach may enable planning that is more defensive and risk-aware than classical methods based solely on the current sensor field of view.

Reference map alignment increases localization robustness to periodic GNSS loss or inconsistency [4], [19], [20]. The difference between the current map and the reference map can indicate not only a localization error, but also an environmental change, the appearance of a new object, or the unreliability of the current GNSS measurement [1]. The three-step alignment procedure (global descriptors, landmarks, local geometric optimization) enables efficient map alignment at varying density and resolution.

Active perception maximizing expected uncertainty reduction relative to energy and computational cost may substantially improve system efficiency under resource constraints [1], [3], [6]. The IR camera may be activated more frequently in low illumination or in regions where prior data indicate the possible presence of a thermal object, while the RGB camera need not be processed by the full network at every frame.

The limitations of the proposed approach include:

1. Implementation complexity - the system requires integration of multiple components (SLAM, segmentation, transformer, prediction, planning) and their efficient synchronization.
2. Computational requirements - despite amortized perception, the spatiotemporal transformer and 4D occupancy forecasting may require significant computational resources.
3. Reference map quality - the effectiveness of reference map alignment depends on the availability and currency of elevation maps or previous flights.
4. Uncertainty calibration - determining calibrated token uncertainty and risk probability requires careful training and validation.
5. Generalization - the model may require fine-tuning for new environments and operational conditions.

Future work should focus on completing and training the learned S4D-TAM model, validating public-dataset converters and pinned external baselines, executing the preregistered H1-H7 mechanism study, and progressing from offline evaluation through SIL, HIL, and controlled real-flight validation. Additional engineering work includes the probabilistic DBN/particle backend, MPPI planning, adaptive octree/LOD storage, active perception, and safety-aware real-time scheduling.

9. Conclusions and Future Research Directions

This paper has proposed S4D-TAM (Semantic 4D Token Attention Map) - a novel navigation system architecture for unmanned aerial vehicles operating in GNSS-degraded environments. The key novelty is the introduction of a persistent memory map composed of semantic 4D tokens, which serves as a unified representation for localization, environment understanding, future scene state prediction, risk estimation, and UAV trajectory planning.

The main conceptual contributions include:

1. Definition of the 4D token combining geometry, semantics, time, uncertainty, and observation history in a single compact representation.
2. Hierarchical system architecture integrating sensor fusion, SLAM, tokenization, spatiotemporal transformer, object memory, reference map alignment, occupancy and risk forecasting, trajectory planning, and active perception.
3. Spatiotemporal attention mechanism accounting for spatial, temporal, and token uncertainty relations.
4. Risk forecasting in partially observable areas based on geometric context, semantics, and observation history.

5. Trajectory planning minimizing length, energy cost, collision probability, predicted environmental risk, and the cost arising from map uncertainty.
6. Active perception maximizing expected uncertainty reduction relative to energy and computational cost.

Seven preregistered mechanism hypotheses have been formulated for semantics, temporal state, calibrated uncertainty, topology, reference-map support, predictive risk, and token lifecycle. Their causal full-vs-H1-H7 study is explicitly separated from the external system comparison against ORB-SLAM3, VINS-Mono, FAST-LIO2, and LIO-SAM.

Future research directions include:

1. Completion and training of the learned S4D-TAM model followed by two-level validation: external system comparison on TartanAir, Blackbird, MARSIM, and AeroVerse, and the preregistered internal full-vs-H1-H7 mechanism study.
2. Computational efficiency optimization through quantization, pruning, knowledge distillation, and dedicated hardware accelerators.
3. Extension with additional sensor modalities (radar, ultrasound, magnetometers) and operational scenarios (multi-robot coordination, long-term missions, dynamic environments).
4. Integration with language models for natural-language-guided navigation and high-level task understanding.
5. Ablation studies of individual system components to identify key factors influencing performance.
6. Validation under real operational conditions on UAV platforms in GNSS-degraded environments.

The proposed S4D-TAM system represents a step toward autonomous UAVs capable of navigating in complex, partially observable environments with limited GNSS signal availability, by exploiting persistent predictive spatial memory in place of repeated analysis of raw sensor streams.

Acknowledgements

The author thanks the reviewers for their valuable comments and the Department of Computer Science and Automation of Rzeszów University of Technology for research support.

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Editorial note: Numbers [12], [14], [16], [18] contain numbering conflicts (the same number used for two different works in different article sections) - these require manual confirmation of full bibliographic data in the final version.

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